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Today
Sinusoidal Positional Encoding ...
Sinusoidal PE Dot Product
Sinusoidal Positional Encoding ...

I want you to collate all the explanations and outputs intelligently into a single output-cover all mentioned there- include each explanations at contextual positions and furnish the output into a single one

Below is a cohesive and comprehensive explanation covering **sinusoidal positional encoding**, clearly organized into logical segments with explicit computations, derivations, and intuitive insights.

✅ Problem Statement:

You're seeking clarity on how the sinusoidal positional encoding formula used in transformers achieves the claim:

"The dot product between PE(pos$_1$) and PE(pos$_2$) depends only on the distance Δ = pos$_2$ – pos$_1$."

Your confusion arises from terms like:

- $\theta_i(pos)$, ω_i
- Dimensions and sin-cos pairs
- Why and how the dot product matches the cos.↓ of the position difference per dimension.

You want a step-by-step example, explicitly computed, to demonstrate this property clearly.

🧑‍🏫 Step-by-Step Example:

1️⃣ Setup and Definitions:

We set the model dimension $d_{\text{model}} = 4$.

Thus, we have $d_{\text{model}}/2 = 2$ pairs of sine and cosine terms.

Pairs are indexed by i:

- **Pair 0:** dimensions 0 (sin), 1 (cos)
- **Pair 1:** dimensions 2 (sin), 3 (cos)

The frequency for each pair is defined as:

$$\omega_i = \frac{1}{10000^{2i/d_{\text{model}}}}$$

2️⃣ Calculate Frequencies (ω_i):

↓

For $d_{\text{model}} = 4$:

- **Pair 0 ($i = 0$):**

$$\omega_0 = 10000^{-2\times0/4} = 10000^{0} = 1$$

- **Pair 1 ($i = 1$):**

$$\omega_1 = 10000^{-2\times1/4} = 10000^{-0.5} = 1/100 = 0.01$$

Thus:

- $\omega_0 = 1$
- $\omega_1 = 0.01$

3️⃣ Compute Positional Encodings:

We explicitly compute $PE(1)$ and $PE(3)$:

🔹 **For Position 1:**

$$PE(1) = [\sin(\omega_0 \cdot 1), \cos(\omega_0 \cdot 1), \sin(\omega_1 \cdot 1), \cos(\omega_1 \cdot 1)]$$

Calculations:

↓

- $\sin(0.01) \approx 0.0099998$, $\cos(0.01) \approx 0.99995$

Thus:

$$PE(1) = [0.84147,\ 0.54030,\ 0.0099998,\ 0.99995]$$

🔹 **For Position 3:**

$$PE(3) = [\sin(\omega_0 \cdot 3), \cos(\omega_0 \cdot 3), \sin(\omega_1 \cdot 3), \cos(\omega_1 \cdot 3)]$$

Calculations:

- $\sin(3) \approx 0.14112$, $\cos(3) \approx -0.98999$
- $\sin(0.03) \approx 0.0299955$, $\cos(0.03) \approx 0.99955$

Thus:

$$PE(3) = [0.14112,\ -0.98999,\ 0.0299955,\ 0.99955]$$

4️⃣ Dot Product of Positional Encodings:

Explicit computation of the dot product $PE(1) \cdot PE(3)$:

↓

$= (0.84147)(0.14112) + (0.54030)(-0.98999) + (0.0099998)(0.0299955) + (0.99995)(0.99955)$

- $0.84147 \times 0.14112 \approx 0.1188$
- $0.54030 \times (-0.98999) \approx -0.5349$
- $0.0099998 \times 0.0299955 \approx 0.0003$
- $0.99995 \times 0.99955 \approx 0.9995$

Sum:

$$0.1188 - 0.5349 + 0.0003 + 0.9995 \approx 0.5837$$

5️⃣ Simplified Dot Product via Trig Identity:

Using the trigonometric identity:

$$\sin(a)\sin(b) + \cos(a)\cos(b) = \cos(b - a)$$

Compute directly in simplified form (dependent only on Δ = 3−1 = 2):

$$= \cos(\omega_0 \cdot 2) + \cos(\omega_1 \cdot 2) = \cos(2) + \cos(0.02)$$

Values:

↓

- $\cos(2) \approx -0.4161$
- $\cos(0.02) \approx 0.9998$

Sum:

$$-0.4161 + 0.9998 \approx 0.5837$$

This matches perfectly with the explicit calculation!

🎯 Key Insight:

The dot product **depends only on the positional distance (Δ)**, and not absolute positions individually:

- $PE(1) \cdot PE(3)$ depends solely on Δ = 2.
- Any positions with the same difference (e.g., $PE(10) \cdot PE(12)$) yield the same dot product.

The formula:

$$PE(pos_1) \cdot PE(pos_2) = \sum_i \cos(\omega_i \cdot (pos_2 - pos_1))$$

clearly illustrates this fundamental property.

↓

🧠 Final Compact ω-form (Clarifying Notation):

Original encoding form:

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{2i/d}}\right), \quad PE(pos, 2i+1) = \cos\left(\frac{pos}{10000^{2i/d}}\right)$$

Converted to the simpler "ω-form":

$$PE(pos, 2i) = \sin(\omega_i \cdot pos), \quad PE(pos, 2i+1) = \cos(\omega_i \cdot pos), \quad \omega_i = \frac{1}{10000^{2i/d}}$$

This form simplifies reasoning about frequency dimensions clearly and intuitively.

✅ How Transformers Leverage This:

- Positional encoding helps transformers differentiate positions based solely on relative distances.
- Attention mechanisms exploit the positional signals to know **"which words are near vs. far"** in a sequence.

↓

🎨 Visualization Offer

Would you now like:

- A visual illustration of how positional encoding components change over positions?
- An additional test (e.g., $PE(5) \cdot PE(7)$) to reinforce your understanding?

This unified explanation combines all your previous queries and insights into one coherent, logically structured presentation to clarify fully how sinusoidal positional encoding achieves its core purpose in transformers.

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